

SAMADRITA ROY

React Native Developer | Mobile & Frontend Engineer

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PROFESSIONAL SUMMARY

React Native Developer with 4+ years of experience at Tata Consultancy Services, delivering enterprise-scale mobile and web applications across aviation, fintech, and government domains. Specialized in React Native, React.js, TypeScript, and cross-platform mobile development with hands-on expertise in iOS/Android app architecture, offline-first patterns, native module integration, and performance optimization. Proven impact delivering production mobile apps supporting 750K+ users, with strong focus on WCAG accessibility, scalable component architecture, and non-functional requirements (NFRs). Actively expanding into Generative AI and LLM-powered mobile applications.

TECHNICAL SKILLS

Mobile Development: React Native (iOS & Android), Expo, React Navigation, Deep Linking, Push Notifications
Languages & Frameworks: JavaScript (ES6+), TypeScript, React.js
State Management & Data: Redux, Redux Toolkit, GraphQL, REST APIs, AsyncStorage, Offline-First Patterns
Mobile Architecture: Cross-Platform Component Architecture, Native Module Integration, Performance Optimization
UI/UX & Accessibility: Responsive Mobile UI, Pixel-Perfect Design, WCAG 2.0/2.1, Cross-Device Compatibility
Testing & Tooling: Unit Testing, Detox (E2E), Storybook, Git, VS Code, Strapi CMS, Xcode, Android Studio
Methods & Practices: Agile/Scrum, Modular Design, Scalable UI Systems, NFR Implementation
Emerging: Generative AI, Prompt Engineering, LLM Integration

PROFESSIONAL EXPERIENCE

React Native Developer / Frontend Developer | Tata Consultancy Services | Kolkata, India *Feb 2022 – Present*

- Engineered and delivered 5+ enterprise-grade mobile and web applications across aviation, fintech, and government sectors, supporting 750K+ users in production.
- Currently building a complete React Native mobile application single-handedly for the Department of Social Protection (DSP), Ireland, targeting production release — covering iOS and Android from architecture to deployment.
- Improved application performance by ~20% through lazy loading, optimized rendering strategies, FlatList optimization, and efficient API consumption patterns.
- Designed and implemented reusable React Native and React component architecture, accelerating development speed by ~70% and improving code maintainability at scale.
- Built scalable mobile and frontend systems using TypeScript, Redux, and GraphQL, ensuring predictable state management and efficient data flow.
- Developed WCAG-compliant accessible UI systems for mobile and web, achieving ~90% audit scores and improving usability for assistive technologies including screen readers.
- Implemented cross-platform responsive UIs ensuring consistent UX across iOS, Android, and web browsers for global users.
- Implemented NFRs including performance, scalability, security, accessibility, and maintainability across multiple production mobile and web applications.
- Collaborated in Agile/Scrum teams, working with backend engineers, designers, and product stakeholders to deliver high-quality releases on schedule.

KEY PROJECTS

Department of Social Protection (DSP), Ireland – Pension Platform (Mobile + Web) *Nov 2024 – Present*

- Sole developer architecting and building the complete React Native mobile application for a government pension platform used by 750K+ users, covering enrollment, opt-in/opt-out workflows, and employer/member portal functionality.
- Designed mobile-first component architecture with React Navigation, ensuring seamless cross-screen navigation and deep linking support.

- Built dynamic multilingual UI using Strapi CMS with a custom BlocksRenderer component to render API-driven content on mobile and web.
- Introduced and implemented internationalization (i18n) for scalable localization across multiple regions, including mobile-specific locale handling.
- Worked extensively with GraphQL APIs for efficient data fetching and structured communication between mobile app and backend.
- Ensured WCAG mobile accessibility compliance and enterprise-grade NFRs across the platform.
- Received recognition from TCS BANCS and TCS Interactive leadership for impactful contributions.

Dibber – Education Platform (Norway)

Jan 2024 – Sep 2024

- Led frontend development for a back-office platform used across 600+ schools in 9 countries, with a focus on scalable architecture and cross-device compatibility.
- Built and maintained a Storybook-based reusable component library, improving UI consistency and developer productivity across teams.
- Implemented Redux-based state management to handle complex application workflows — patterns directly transferable to React Native state architecture.
- Applied WCAG 2.0 accessibility standards for inclusive user experience; implemented performance optimization using lazy loading and render optimization.
- Received direct client appreciation and recognition from Global Head of TCS Interactive.

OMSA, Air India Express & Standard Bank – Enterprise Platforms

Feb 2022 – Nov 2024

- Delivered frontend solutions across insurance (OMSA), airline booking (Air India Express), and banking (Standard Bank) domains.
- Built API-driven UI systems, reusable React components, booking workflows, business logic integrations, and responsive pixel-perfect interfaces for enterprise-scale platforms.

PERSONAL PROJECTS

AI Agent (GitHub)

- Developed a GenAI-powered AI agent leveraging LLaMA3 via Ollama, enabling intelligent task execution through prompt-driven workflows.
- Integrating AI workflows with React-based frontend systems to enable real-time AI-driven UI interactions — exploring mobile AI integration via React Native.

EDUCATION

M.Sc in Electronic Science | Calcutta University | Kolkata, India

Aug 2018 – Oct 2020

ACHIEVEMENTS & RECOGNITION

- Recognized by Global Head of TCS Interactive for outstanding contributions on Dibber platform (Norway).
- Awarded "Star of the Month" (2x) for high-impact performance and delivery on the DSP Ireland government project.
- Appreciated by TCS BANCS and TCS Interactive leadership for enterprise-scale delivery and mobile application development.
- Sole React Native developer on a 750K+ user government platform — from architecture to production release.